

PATIENT EDUCATION · NEPHROLOGY

Phosphorus Control in CKD

Protecting your bones and blood vessels from high phosphorus — understanding bone disease prevention and **vascular calcification** in chronic kidney disease.



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3.5–5.5

mg/dL Normal Range

>5.5

mg/dL Hyperphosphatemia

Bone Disease

Primary Risk

Binders

Key Treatment

CKD impairs phosphorus excretion — what your kidneys cannot remove builds up in blood and damages bones, vessels, and heart.

1 Why High Phosphorus Is Dangerous in CKD

Complication	What Happens	Consequence
Secondary Hyperparathyroidism	High phosphorus suppresses calcitriol → PTH rises to pull calcium from bones	Bones weaken, fracture risk rises
Renal Osteodystrophy	Chronic PTH excess dissolves bone mineral; bones become soft and deformed	Pain, fractures, bone deformity
Vascular Calcification ("Stone Heart")	Calcium-phosphorus product >55 precipitates in arterial walls and heart valves	Stiff arteries, heart attack, stroke
Itching (Pruritus)	Calcium-phosphate crystals deposited in skin trigger nerve irritation	Severe generalized itching, poor sleep
Cardiovascular Death	Calcified arteries + high PTH + heart valve calcification all compound risk	Leading cause of death in dialysis patients

2 Phosphorus Targets by CKD Stage

CKD Stage	eGFR (mL/min)	Phosphorus Target	Key Action
Stage 1–2	≥60	Normal (3.5–5.5 mg/dL)	Routine monitoring; avoid phosphate-heavy processed foods
Stage 3a–3b	30–59	3.5–5.5 mg/dL	Dietary counseling; check PTH; reduce processed foods
Stage 4	15–29	3.5–5.5 mg/dL	Phosphate binders may be needed; strict dietary restriction
Stage 5 / Pre-dialysis	<15	3.5–5.5 mg/dL	Phosphate binders required; aggressive dietary restriction
Hemodialysis	On HD	3.5–5.5 mg/dL	Binders with every meal; 3× weekly HD removes ~800–1000 mg

3 Organic vs Inorganic Phosphorus — Not All Phosphorus Is Equal

Organic Phosphorus (from Food)

Naturally bound to protein in animal and plant foods

Absorption: 40–60% absorbed from gut

Animal protein sources: meat, fish, poultry, eggs, dairy — phosphorus is protein-bound; moderate absorption even without binders.

Plant sources: legumes, nuts, seeds — phosphorus is bound to phytate; **only 20–40% absorbed** (humans lack phytase enzyme). Vegetarians absorb less.

Examples: tilapia, bangus, chicken, tofu, monggo, mani, pili nuts

Inorganic Phosphorus (from Additives)

Free phosphate salts added to processed foods — no protein binding

Absorption: 80–100% absorbed — far more dangerous

Why so dangerous: Not bound to protein → immediately absorbed in gut → bypasses normal regulatory steps → spikes serum phosphorus rapidly.

✗ Found in: cola drinks (phosphoric acid), fast food, processed meats (hotdog, longganisa), instant noodles, bread improvers, cheese spreads.

Label keywords: phosphoric acid, sodium phosphate, pyrophosphate, polyphosphate — anything with "PHOS"

⚠ Critical Warning — Phosphate Additives

Phosphate additives in processed foods (cola drinks, fast food, processed meats) are absorbed at 80–100% — they are far more dangerous than natural food phosphorus. A single can of cola adds more bioavailable phosphorus than a serving of fish. Eliminate these first.

CKD-MBD CASCADE — HOW HIGH PHOSPHORUS DAMAGES BONES AND HEART

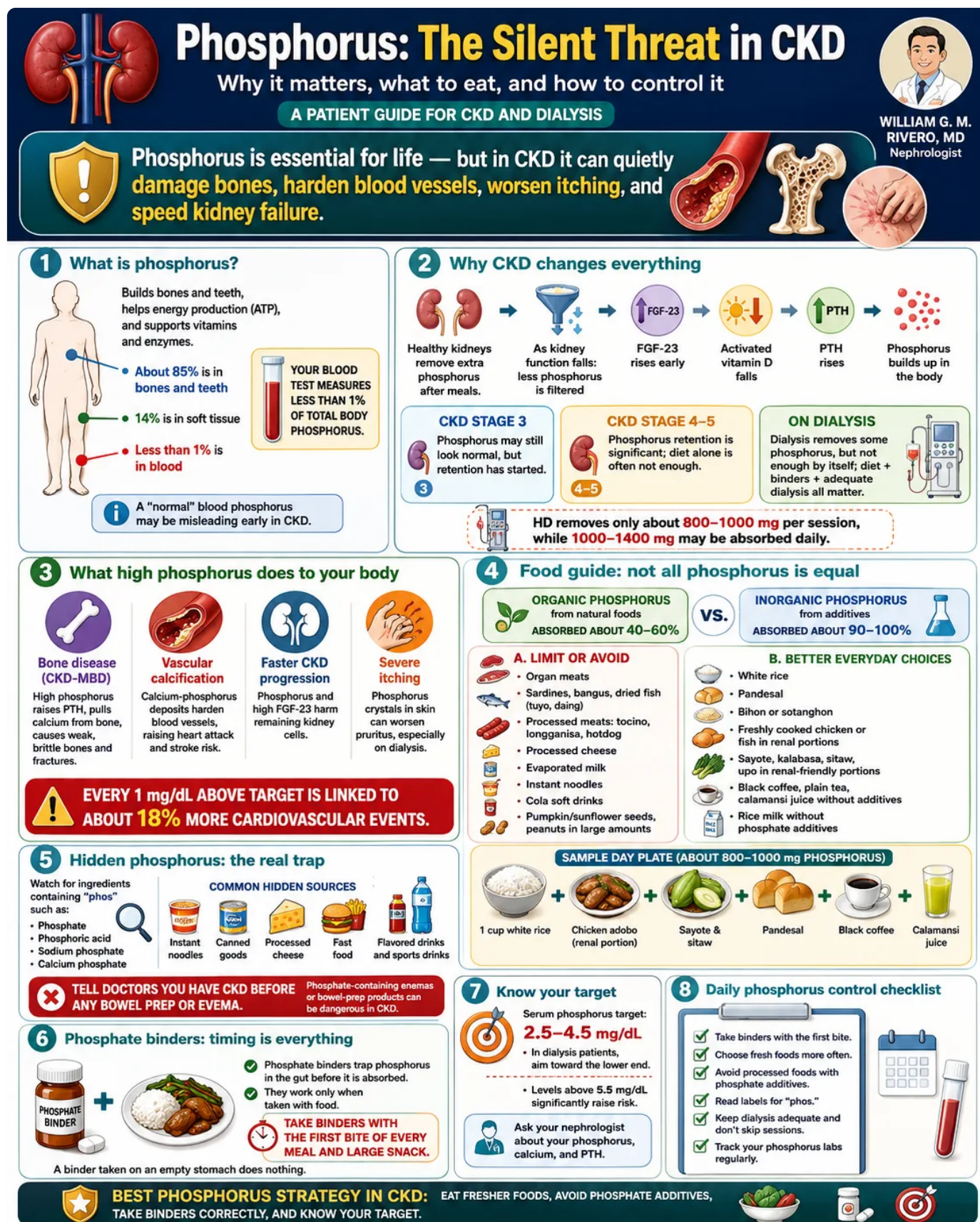


Fig. 1 — The CKD-Mineral and Bone Disorder (CKD-MBD) cascade: impaired phosphorus excretion → elevated serum phosphorus → suppressed calcitriol → secondary hyperparathyroidism → renal osteodystrophy and vascular calcification. Controlling dietary phosphorus and using phosphate binders breaks this cascade early, preventing bone fractures and cardiovascular events.

Food (Filipino name)	Standard Serving	Phosphorus	Notes for CKD Patients
HIGH — >200 MG/SERVING · LIMIT OR AVOID			
Atay ng manok / chicken liver	100 g	350 mg	Organ meats — very high phosphorus. Avoid in CKD 4–5 and dialysis.
Kidney organ meat	100 g	240 mg	All organ meats are very high in phosphorus. Eliminate from CKD diet.
Evaporated milk (gatas de baka)	½ cup (125 mL)	260 mg	Common in Filipino cooking — substitute with non-dairy creamers (check label for phosphate additives).
Milk (fresh / pasteurized)	1 cup (250 mL)	230 mg	Limit dairy milk in CKD 4–5. Rice milk or small amounts preferred.
Processed cheese (quick melt)	1 slice (30 g)	200 mg	Processed cheese has added inorganic phosphate — absorbed 80–100%. Avoid.
Monggo / mung beans (boiled)	½ cup cooked	200 mg	Plant phosphorus (phytate-bound) — 20–40% absorbed. Limit in CKD 4–5; boil and discard water first.
Utong / bataw (cowpeas/hyacinth beans)	½ cup cooked	180 mg	Similar to monggo — phytate-bound phosphorus. Moderate in CKD 3; limit in CKD 4–5.
Cola drinks (Coke, Pepsi)	1 can (355 mL)	Additive	Phosphoric acid — inorganic, 80–100% absorbed. Eliminate completely in all CKD stages.
Pili nuts	¼ cup	140 mg	High in phosphorus. Limit in CKD 4–5. Small portions (10–15 nuts) in earlier CKD.
Instant noodles (Mami, Lucky Me)	1 pack	150 mg	Contains phosphate additives in seasoning — inorganic. Avoid in CKD. Use plain noodles without flavoring sachets.
Mani / peanuts	¼ cup	130 mg	Plant-bound phosphorus — lower absorption. Limit to small handful in CKD 4–5.
Hotdog / longganisa	1 piece	120 mg	Contains added inorganic phosphates as preservatives. Avoid — high sodium and phosphate additives.
MODERATE — 100–200 MG/SERVING · USE CAREFULLY			
Pork (karne, any cut)	100 g cooked	195 mg	High-quality protein needed in CKD — take phosphate binder with meal. Boil and discard broth to reduce.
Chicken (manok)	100 g cooked	180 mg	Good protein choice. Always take binder with chicken meals. Avoid skin (additional fat and phosphorus).
Tilapia	100 g cooked	175 mg	Excellent lean protein for CKD. Take binder with each fish meal. Preferred over processed fish products.
Bangus (milkfish)	100 g cooked	170 mg	Good fresh fish choice. Take binder with meal. Avoid canned bangus (may have phosphate additives).
Cheese (natural, kesong puti)	1 slice (30 g)	140 mg	Natural cheese is better than processed — no added inorganic phosphate. Still limit in CKD 4–5.
Tofu (firm)	100 g	120 mg	Plant-based phosphorus — lower absorption. Good protein substitute for meat in CKD. Calcium-set tofu preferred.
Egg (itlog)	1 large	95 mg	Excellent protein source. Take binder with meals. Egg white only has very low phosphorus (5 mg) — use if phosphorus is very elevated.
Rice (kanin, white, cooked)	1 cup cooked	68 mg	Lowest phosphorus staple grain. Safe at normal portions in all CKD stages. Best carb choice for dialysis.
Sweet potato / kamote	1 medium boiled	50 mg	Low phosphorus for a starchy root crop. Boil and discard water. Moderate potassium — check with dietitian.
Pandesal / bread	1 piece	40 mg	Commercial bread may contain phosphate dough conditioners — check label. Homemade pandesal preferred.
LOW — <100 MG/SERVING · GENERALLY SAFE			
Kangkong (water spinach)	1 cup cooked	20–30 mg	Excellent low-phosphorus vegetable. Safe for all CKD stages. Good daily leafy green.
Pechay (bok choy)	1 cup cooked	25 mg	Very safe. Low phosphorus and potassium. Ideal for CKD and dialysis patients.
Sayote (chayote)	1 cup cooked	30 mg	Low phosphorus, low potassium. Good bulk vegetable in sinigang and soups.
Ampalaya (bitter melon)	1 cup cooked	35 mg	Low phosphorus. Also benefits blood sugar control in diabetic CKD patients.
Okra	1 cup cooked	40 mg	Excellent vegetable for CKD — low phosphorus, low potassium, high fiber. Eat daily.
Upo (bottle gourd)	1 cup cooked	20 mg	Very low phosphorus. Versatile in Filipino cooking. Safe for all CKD stages.
Cassava / kamoteng kahoy	1 cup cooked	28 mg	Low phosphorus starchy root. Boil thoroughly. Limit canned or sweetened cassava products.
Gabi (taro)	1 cup cooked	40 mg	Low phosphorus. Moderate potassium — boil and drain. Good carbohydrate for CKD.
Sago (tapioca pearls)	½ cup cooked	Negligible	Near-zero phosphorus and potassium. Safe starchy addition to drinks and soups.
Banana (saging, ripe)	1 medium	22 mg	Low phosphorus but moderate-to-high potassium. Limit to ½ banana in CKD 4–5 / dialysis.

Food (Filipino name)	Standard Serving	Phosphorus	Notes for CKD Patients
Papaya (hinog na papaya)	1 cup chunks	20 mg	Very low phosphorus and potassium. Good daily fruit for CKD patients.
Calamansi	5 pieces	5 mg	Negligible phosphorus. Good flavoring to replace salt. Use freely.
Cooking oils (coconut, canola)	1 tbsp	0 mg	No phosphorus. Preferred over processed fats. Use in moderation for calorie needs.

Plant phosphorus (phytate-bound) absorbed 20–40%; animal phosphorus absorbed 40–60%; inorganic additive phosphorus absorbed 80–100%. Values from FNRI Philippine Food Composition Tables 2023. All values approximate — serving size, preparation method, and cooking affect final content.

Phosphate Binders — Types, Doses, and How to Use Them				
Binder	Brand (Philippines)	Typical Dose	Take With	Notes
Calcium carbonate	Caltrate, local generics	500–1500 mg	With each meal	Also used as calcium supplement — watch for hypercalcemia in dialysis. Avoid if calcium is already high.
Sevelamer HCl	Renagel	800 mg	With each meal	Non-calcium, non-aluminum binder. Preferred in patients with vascular calcification or high calcium. Also lowers LDL.
Lanthanum carbonate	Fosrenol	500–1000 mg	With each meal	Chew thoroughly — do not swallow whole. Very effective. No calcium load. More expensive.
Calcium acetate	PhosLo	667 mg	With each meal	Less elemental calcium than carbonate per dose — slightly lower hypercalcemia risk. Still monitor calcium.
Aluminum hydroxide	Amphojel	Short-term only	With meals	Very effective but causes aluminum toxicity with long-term use. Only for acute short-term control. Not for maintenance.

 **Binder Timing Is Everything**

Phosphate binders **MUST be taken WITH meals** — not before or after. They bind dietary phosphorus in the gut before it is absorbed. Skipping even one meal dose allows that meal's entire phosphorus to be fully absorbed. A binder taken 1 hour after a meal is largely useless.

The Additive Phosphorus Problem — Reading Food Labels

Dangerous Additives to Avoid (look on ingredient list):

- **Phosphoric acid** — in all cola drinks, some energy drinks
- **Sodium phosphate, sodium tripolyphosphate** — in processed meats, canned tuna, fast food chicken
- **Potassium phosphate** — in bread improvers, sports drinks
- **Calcium phosphate** — in some fortified foods, baking powder
- **Pyrophosphate** — in instant noodle seasoning, processed cheese
- **Polyphosphate** — in deli meats, seafood marinades, frozen food

Commonly Found In (Philippine market):

- Cola drinks (Coke, Pepsi, RC Cola)
- Fast food fried chicken (KFC, Jollibee, Chowking)
- Processed meats (hotdog, spam, corned beef, tocino)
- Instant noodles (Lucky Me, Mami, Quickchow)
- Bread improvers (commercial sliced bread)
- Processed / quick-melt cheese (Eden, similar brands)

Simple Label Rule

Anything with **"PHOS"** anywhere in the ingredients list — avoid. This one rule eliminates the most dangerous phosphorus sources.

Cooking Techniques to Reduce Phosphorus Content

Technique	Phosphorus Reduction	How to Do It
Boiling meat	30–55% reduction	Boil chicken, pork, or fish in large volume of water. Always discard cooking water — phosphorus leaches out. Do not use broth for soups in CKD.
Boiling vegetables	30–50% reduction	Boil vegetables 10–15 minutes in large water, drain and discard water. Re-cook in fresh water for soups. Reduces both phosphorus and potassium.
Soaking legumes	20–35% reduction	Soak dried monggo, beans, or lentils overnight in cold water. Discard soaking water completely. Boil in fresh water. This removes much of the phytate-bound phosphorus.
Fresh over canned	Avoid additives	Canned tuna, sardines, and salmon often contain sodium phosphate as a preservative. Buy fresh or frozen fish — no additives.
Avoid roasting / grilling	Concentrates phosphorus	Roasting, grilling, and frying concentrate minerals as water evaporates. Boiling is always the best phosphorus-reduction cooking method for CKD.

Symptoms of Hyperphosphatemia — Often Silent Until Advanced		
Symptom	Mechanism	CKD Stage When Seen
Often No Symptoms Early	Phosphorus rises silently — most patients feel nothing until levels are very high	CKD 3–4 (subclinical)
Itching (pruritus)	Calcium-phosphate microcrystals deposited in skin — activates itch receptors	CKD 4–5, dialysis
Bone pain, joint pain	Renal osteodystrophy — PTH-driven bone turnover causes aching	CKD 4–5, dialysis
Eye redness, "red eye"	Calcium-phosphate deposits in conjunctiva — calcific conjunctivitis	Advanced CKD / dialysis
Pathologic fractures	Severe renal osteodystrophy — bones break with minimal trauma	Advanced / long-term CKD
Calciophylaxis (skin necrosis)	Ca×P product precipitates in small blood vessels → skin ischemia → necrosis	Emergency — dialysis patients

Lab Monitoring Targets (KDIGO 2024)		
Lab / Parameter	CKD 3–4 Target	Dialysis Target
Serum phosphorus	3.5–5.5 mg/dL	3.5–5.5 mg/dL
PTH (parathyroid hormone)	Stage 3: 35–70 pg/mL Stage 4: 70–110 pg/mL	2–9× upper normal (150–600 pg/mL)
Serum calcium	8.5–10.5 mg/dL	8.5–10.2 mg/dL (avoid hypercalcemia)
Ca × P product	<55 mg ² /dL ²	<55 mg ² /dL ² (strict)
Vitamin D (25-OH)	>30 ng/mL (replete if <20)	Use active calcitriol — not regular D3
Alkaline phosphatase	Normal (<130 U/L)	<130 U/L (elevated = active bone disease)

Special Situations			
Dialysis Patients Each hemodialysis session removes only ~800–1000 mg phosphorus. A typical Filipino diet delivers 1200–2000 mg/day. Three sessions per week cannot compensate for a high-phosphorus diet — binders and dietary restriction are both essential.	Vegetarian / Vegan CKD Plant phosphorus is phytate-bound — only 20–40% is absorbed versus 40–60% for animal sources. Vegetarians and vegans with CKD have some latitude with legumes (monggo, tofu). Still avoid phosphate-additive processed foods. Soaking and boiling legumes before eating reduces phosphorus further.	Children with CKD Children need adequate phosphorus for bone growth — targets are stricter and individualized. Never restrict phosphorus in children without pediatric nephrology guidance. Formula and dairy choices must balance CKD phosphorus control against growth needs. PTH monitoring is especially important.	KA Supplement Users Each ketoanalogue (KA) tablet contains calcium carbonate — this contributes to calcium load. When taken with meals, KA tablets also act as mild phosphate binders. Monitor calcium × phosphorus product closely. Do not add separate calcium-containing binders without physician review if already on KA.

Calciophylaxis — Life-Threatening Emergency

Calciophylaxis presents as **painful skin lesions with necrosis**, typically on the abdomen, thighs, or buttocks. The skin appears mottled, then develops black/purple ischemic patches. It usually occurs with very high calcium × phosphorus product (>70 mg²/dL²). Mortality is very high. **Go to the ER immediately** if painful, non-healing skin lesions appear — especially in dialysis patients with uncontrolled phosphorus.

References: KDIGO 2017 Clinical Practice Guideline Update for the Diagnosis, Evaluation, Prevention, and Treatment of CKD-MBD · KDIGO 2024 CKD Clinical Practice Guideline · Uribarri J, et al. "Hidden sources of phosphorus in the typical American diet." Semin Dial. 2003 · FNRI Philippine Food Composition Tables 2023 · Sullivan C, et al. "Effect of food additives on hyperphosphatemia among patients with end-stage renal disease." JAMA. 2009.